Predicting Adolescents' Future Smoking Behavior through Anti-Smoking Ad Exposure: A Structural Equation Modeling Approach Based on the Theory of Planned Behavior

Source: Freepik. Say "No" to Smoking Today - World No Tobacco Day. Freepik. https://www.freepik.com

by

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Introduction & Rationale

THE PROBLEM

- Over 2.22 million High school and Middle school students are current smokers.
- Adolescents = key demographic (initiation < age 18).
- Smoking = leading cause of preventable death (490,000+ deaths/year in U.S. from tobacco-related illnesses)

INDUSTRY INFLUENCE

- $8.6 B spent/ year on tobacco marketing.
- Media glamorization normalizes smoking.

KNOWLEDGE GAP

- Do anti-smoking advertisements influence adolescents' perceptions or behaviors?

PREVENTION STRATEGY

- Anti-smoking ads aim to:
  - ✓ Raise risk perception.
  - ✓ Strengthen peer disapproval.
  - ✓ Empower refusal confidence.

Research Objective & Hypotheses

- **Objective**
  - Examine how anti-smoking ads and constructs within TPB theory predict smoking intentions and behavior

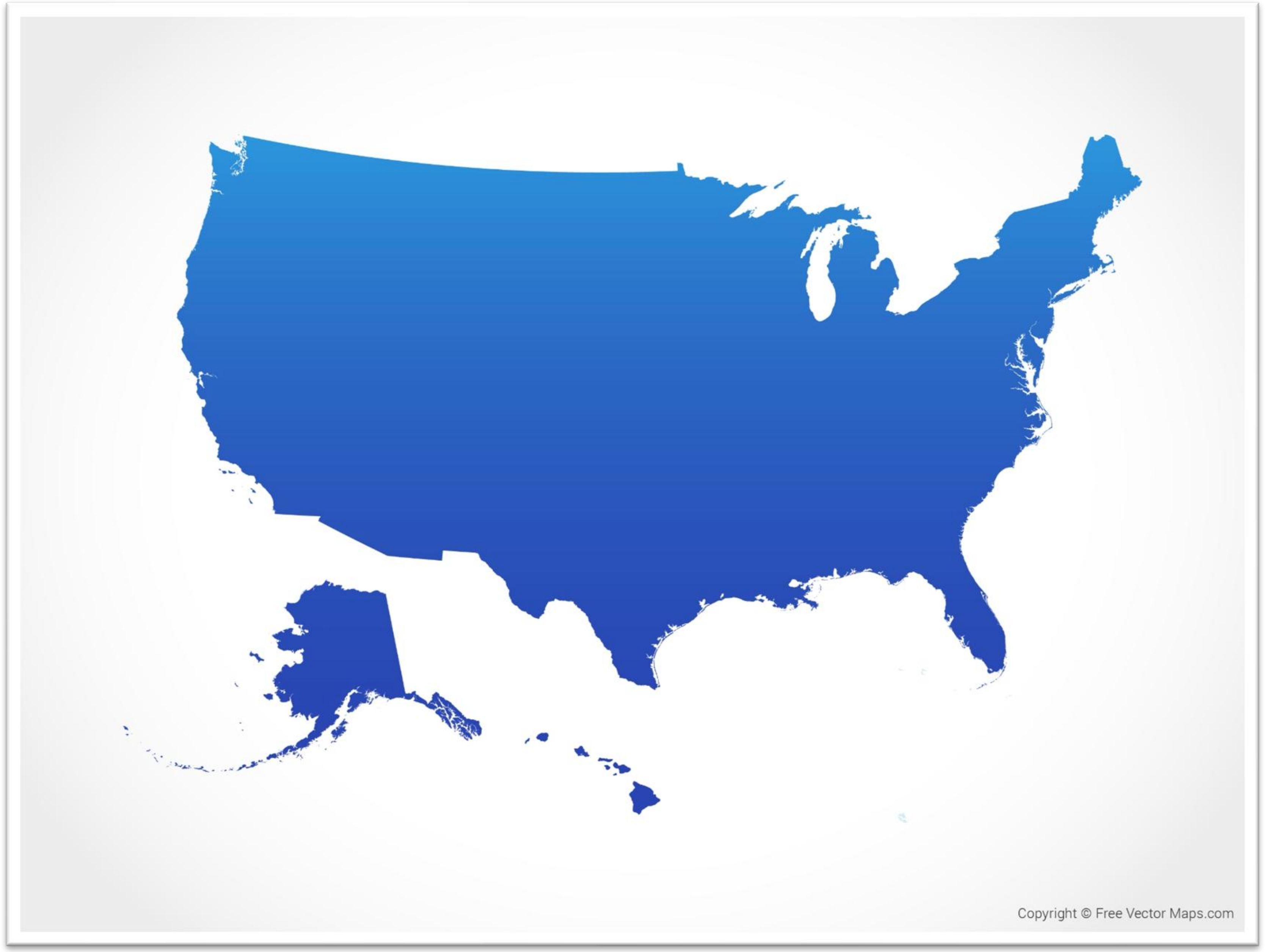
- **Hypothesis**
  - H_0: Anti-smoking advertisements have no direct or indirect correlation on adolescents' intentions to use tobacco, nor do they significantly influence attitudes, perceived social norms, or perceived behavioral control in predicting current smoking behavior.
  - H_1: Anti-smoking advertisements have a direct and indirect correlation on adolescents' intentions to use tobacco. These advertisements shape key psycho-cognitive factors—such as attitudes, perceived social norms, and perceived behavioral control—which, in turn, influence future smoking intentions and affect current smoking behavior.

Data & Methods

- **Data Source**
  - Dataset: Monitoring the Future (MTF), 2023
  - Design: Cross-sectional, nationally representative
  - Sample: N ≈ 14,000 adolescents (8th & 10th grade)
  - Selection: 3-stage stratified sampling
  - Ethics: No identifiable info on respondents.

- **Variables & Constructs**
  - Latent Constructs (via CFA):
  - Attitudes (ATT) – 3 items
  - Injunctive Norms (INJ) – 2 items
  - Descriptive Norms (DES) – 2 items
  - Perceived Behavioral Control (PBC) – 1 item

- **Observed Variables**
  - Intention (INT) – Future smoking plan
  - Behavior (SMK) – Past 30-day smoking
  - Anti-Smoking Ad Exposure (ADS) – Binary
  - Controls: Gender, Race, Grade, Parental Presence



Source: Map pf the United States of America - Blue. US-EPS-02-4001. Free Vector Maps July 2025.

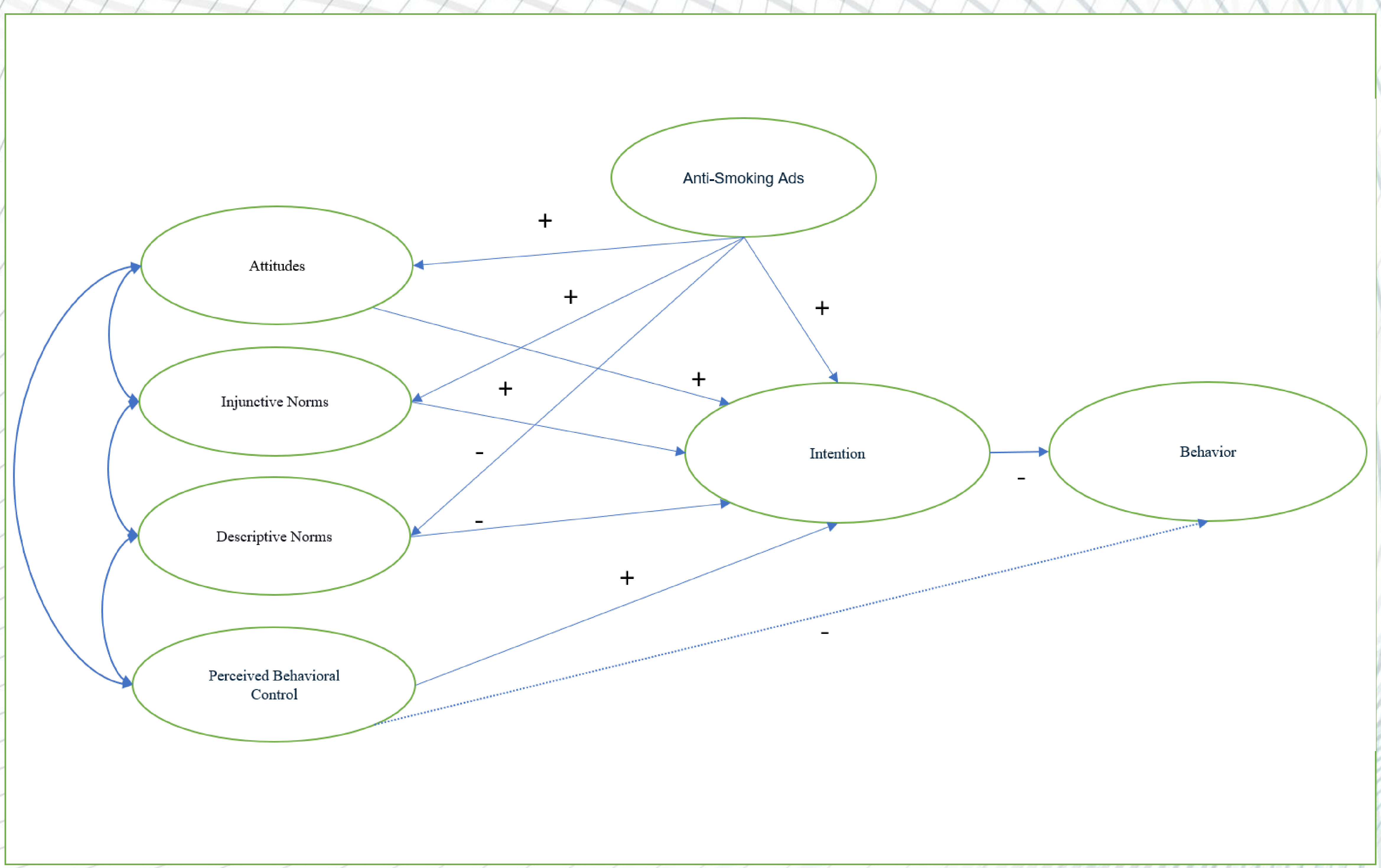
Measurement Model: Constructs and Indicators

Construct	Indicators	Survey Question (Paraphrased)	Scale
Attitudes	att1, att2, att3	Perceived risk from using tobacco	1 = No risk → 4 = Great risk
Injunctive Norms	inj1, inj2	Friends' disapproval of smoking	1 = Not disapprove → 3 = Strongly disapprove
Descriptive Norms	des1, des2	Friends' tobacco usage	1= None → 5=All
Perceived Control	pbc	Ability/confidence to avoid smoking	1 = Not at all → 4 = Very
Intention	int	Intention of smoking in the next five years	1 = Definitely not → 4 = Definitely yes
Behavior	smk	Current smoking behavior	0 = No / 1 = Yes
Ad Exposure	ads	Exposure to anti-smoking ads	0 = No / 1 = Yes

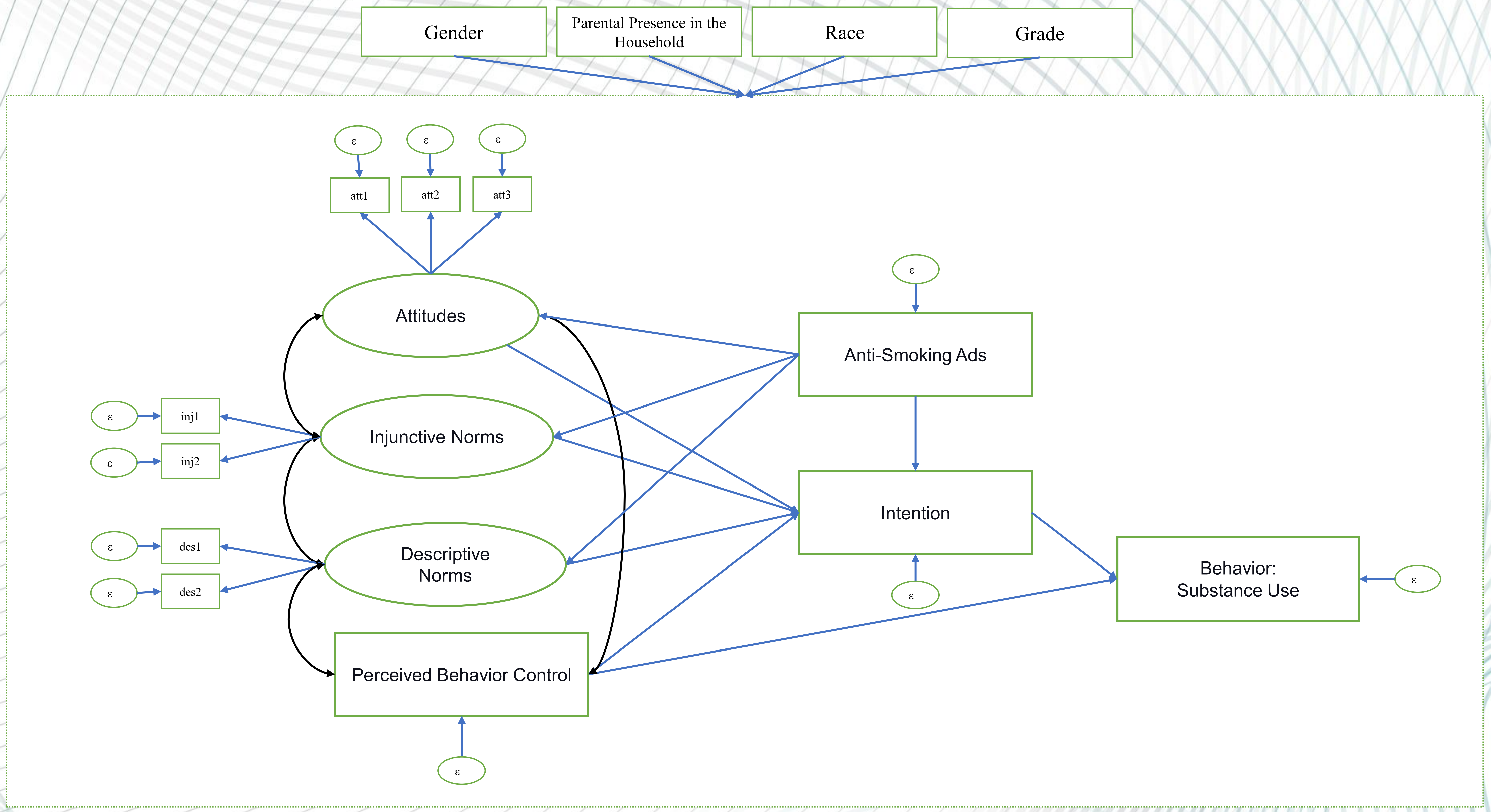
Note: All indicators are ordinal categorical variables from a nationally representative questionnaire.

Conceptual Model

Theoretical Framework: Theory of Planned Behavior with Ads & Demographics

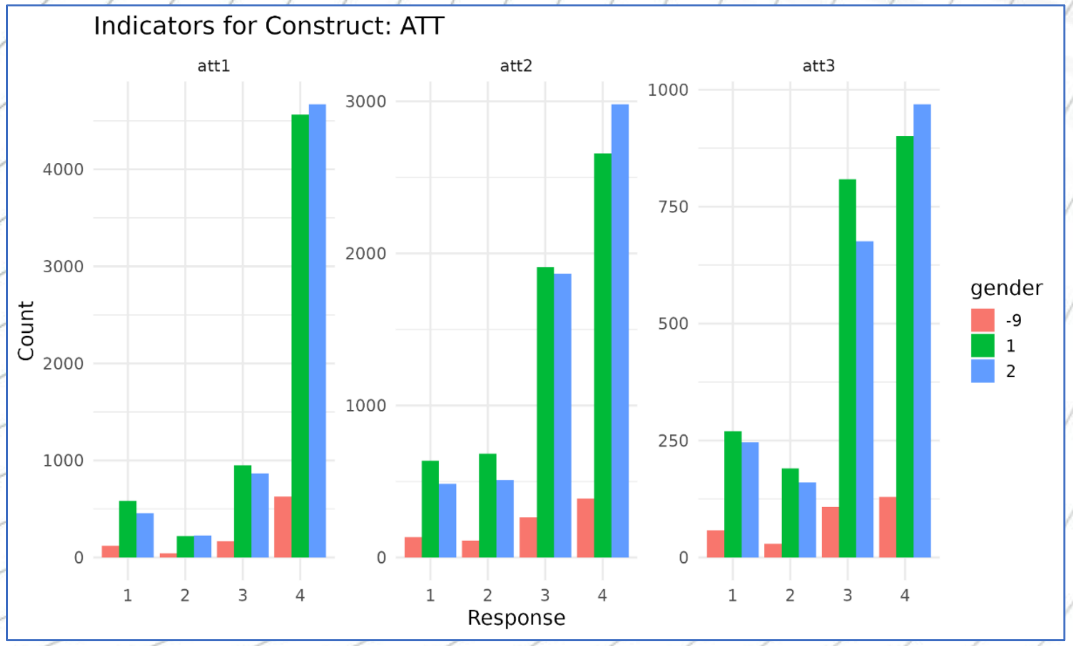


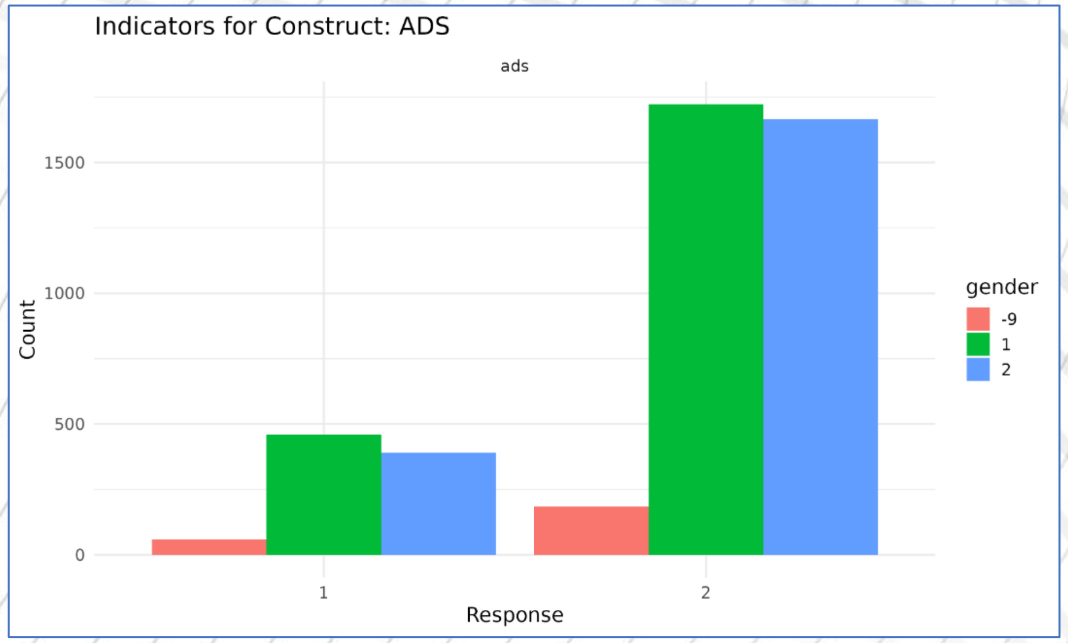
Hypothesized Model

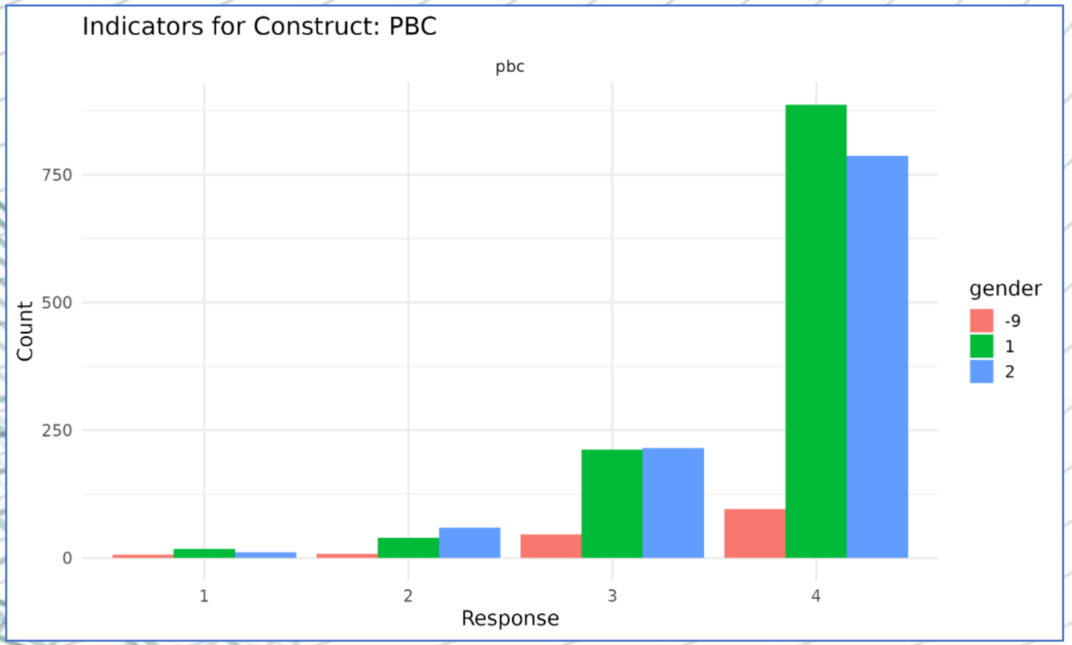


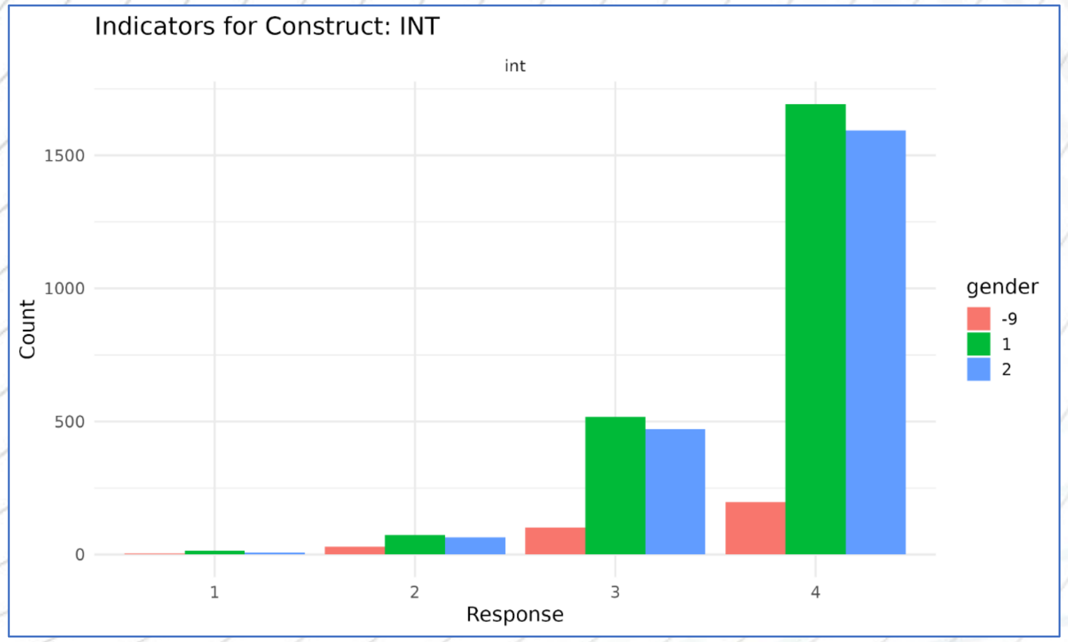
Latent & Observed Constructs with Demographics Visualizer

Interact Live with Shiny App for Data Visualization: mye-chow.shinyapps.io/rstudio









Note: Figures showing some examples of the plots created using the R Shiny App.

Purpose of the Interactive Tool:

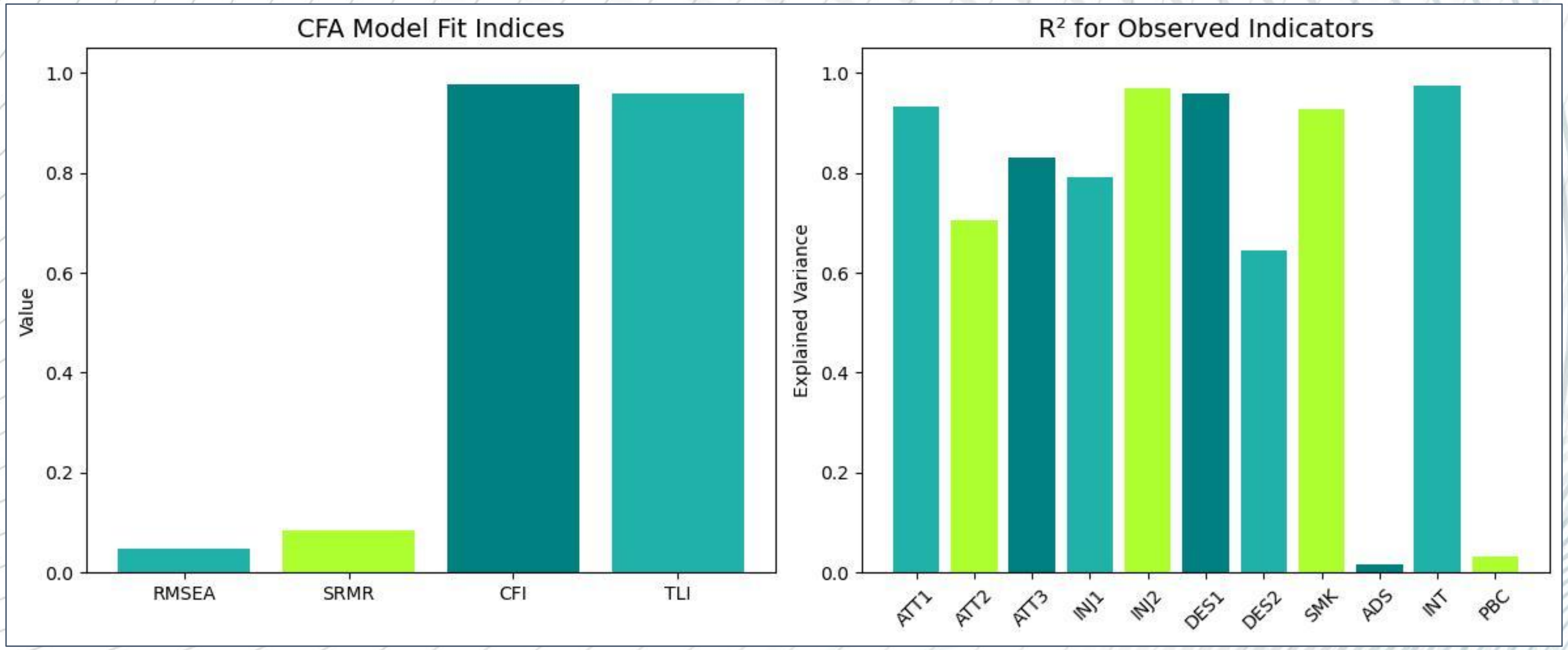
The Shiny App enables real-time exploration of the Theory of Planned Behavior (TPB) theory, allowing users to visualize the impact of anti-smoking ad exposure and psycho-cognitive predictors across demographic subpopulations.

What the Visualization Shows:

Each TPB construct is displayed through its observed indicators, stratified by demographic groups such as gender, grade, race, and parental presence. Users can interactively select constructs and compare response distributions across groups.

Measurement Validity: CFA

- **CFI** (Comparative Fit Index) = **0.975**
- **TLI** (Tucker-Lewis Index) = **0.952**
- **RMSEA** (Root Mean Square Error of Approximation) = **0.036**
- **SRMR** (Standardized Root Mean Square Residual) = **0.090**



Note: Figure created with Jupyter Notebook using Python script

Factor Analysis of Latent Constructs

- Purpose
  - Validate the latent constructs (Attitudes, Injunctive Norms, Descriptive Norms) using CFA prior to SEM path analysis.
  - Assess how well observed indicators reflect their theoretical factors under the TPB framework.

- Key Findings
  - Strong Internal Consistency:
    - ❑ Attitudes (ATT): CR = 0.93, AVE = 0.82
    - ❑ Injunctive Norms (INJ): CR = 0.94, AVE = 0.59
    - ❑ Descriptive Norms (DES): CR = 0.89, AVE = 0.53.
- Interpretation
  - Latent constructs are psychometrically strong and have great internal alignment.

SEM Path Analysis

- Confidence in Resisting Smoking was the strongest predictor for Intention to Not Smoke for 5 Years (β = 0.817)
- Intention to Not Smoke for 5 Years→ Smoking status (β = –1.384)
- Exposure to Anti-Smoking Ads→ Intention to Not Smoke for 5 Years: Unexpected negative β = -0.253
- Confidence in Resisting Smoking → Smoking status: Unexpected positive β = 0.738

Path	β	S.E.
Confidence in Resisting Smoking → Intention to Not Smoke for 5 Years	0.817	0.018
Risk Perceptions of Using Tobacco → Intention to Not Smoke for 5 Years	0.201	0.029
Peer Disapproval of Smoking → Intention to Not Smoke for 5 Years	0.432	0.035
Prevalence of Peers Using Tobacco → Intention to Not Smoke for 5 Years	-0.343	0.031
Intention to Not Smoke for 5 Years→ Smoking status	-1.384	0.102
Exposure to Anti-Smoking Ads→ Intention to Not Smoke for 5 Years	-0.253	0.053
Confidence in Resisting Smoking → Smoking status	0.738	0.102
Exposure to Anti-Smoking Ads → Risk Perceptions of Using Tobacco	0.347	0.030
Exposure to Anti-Smoking Ads → Peer Disapproval of Smoking	0.384	0.034
Exposure to Anti-Smoking Ads → Prevalence of Peers Using Tobacco	-0.145	0.039

Demographic Insights

❖ **Parental Presence**

- → Exposure to Anti-Smoking Ads (β = 0.09, p = 0.003).

❖ **Grade level (10th vs. 8th)**

- → Risk Perceptions of Using Tobacco (β = 0.128, p = 0.00)
- → Peer Disapproval of Smoking (β = -0.047, p = 0.000)
- → Prevalence of Peers Using Tobacco (β = 0.114, p = 0.000)
- → Intention to Not Smoke for 5 Years (β = 0.110, p = 0.003). → Smoking Status (β = 0.058, p= 0.000)

❖ **Males**

- → Risk Perceptions of Using Tobacco (β = –0.059, p = 0.00)
- → Prevalence of Peers Using Tobacco (β = –0.059, p = 0.006).

❖ **For Black adolescents**

- → Risk Perceptions of Using Tobacco (β = -0.057, p = 0.000)
- → Prevalence of Peers Using Tobacco (β = -0.109, p = 0.000)
- → Intention to Not Smoke for 5 Years (β = -0.102, p = 0.021)
- → Confidence in Resisting Smoking (β = 0.168, p = 0.00).

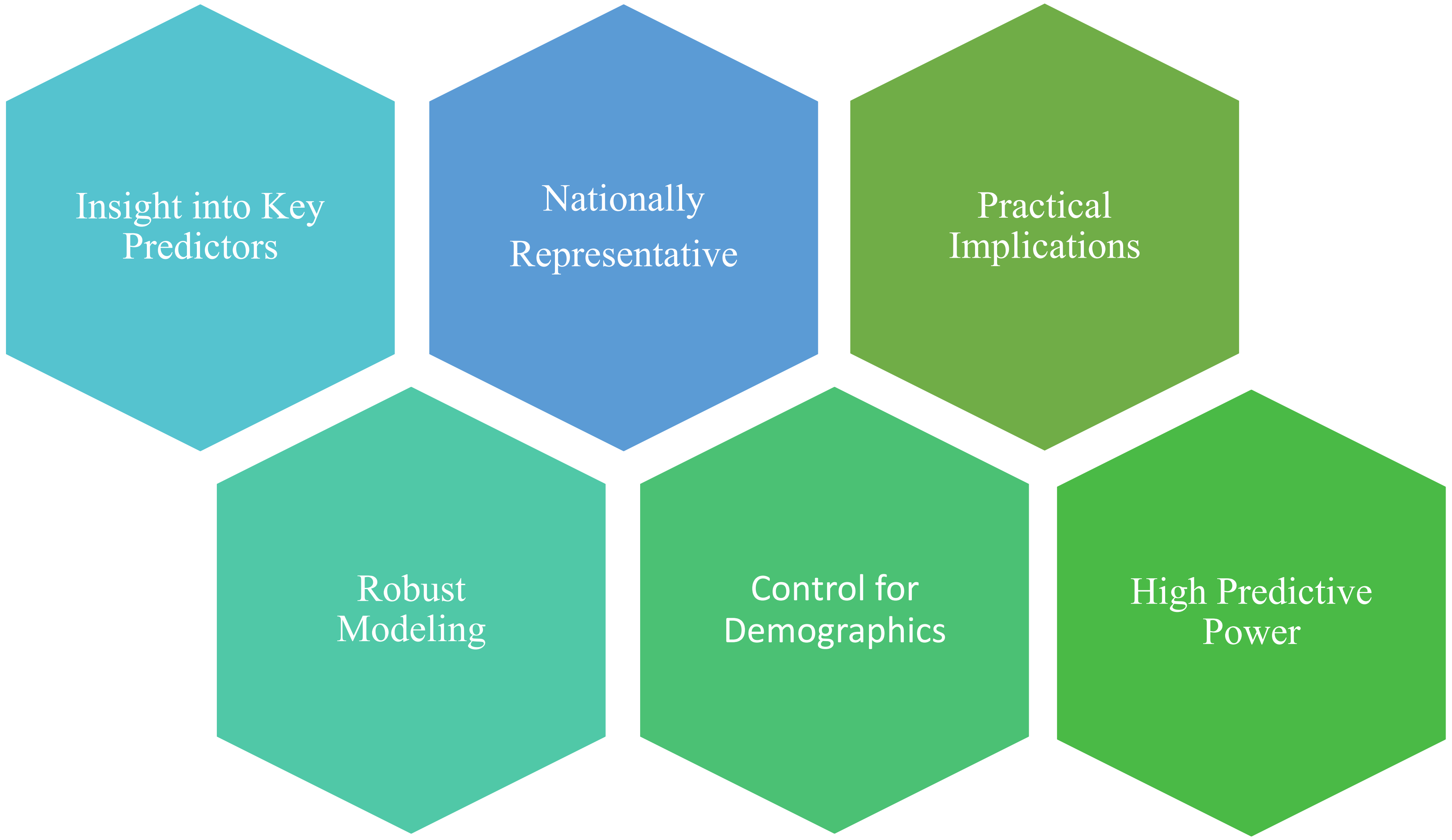
❖ **For Hispanic adolescents**

- → Risk Perceptions of Using Tobacco (β = –0.072, p = 0.000)
- → Smoking status (β = –0.194, p = 0.003).

Discussion

- SEM results rejected the null hypothesis (H_0) and statistically confirmed that anti-smoking ads play a significant role in shaping adolescents' smoking-related beliefs and intentions.
- Strategic targeting and early intervention are essential because adolescents are the future adults of tomorrow.
- Anti-smoking ads are insufficient if not paired with adolescents perceived attitudes, norms, and behavior control to predict future smoking behavior.
- Adolescents' sense of perceived behavior control truly protects them against future smoking, more than anti-smoking ads and peer disapproval.

Strength of the Research



Limitations of the Research

Cross-sectional Design	Not able to predict actual future behavior or infer causality due to lack of longitudinal data for the respondents' intentions.
Modest Intervention Impact	Anti-smoking ad exposure had a minimal but negative influence on future smoking avoidance intentions.
Measurement Simplifications	Injunctive and Descriptive Norms measured using only two indicators each.
Possible Social Desirability Bias	Self-reported behaviors and intentions may reflect socially desirable responses rather than actual attitudes or behaviors.
Low Smoking Prevalence	Only ~1% reported recent smoking, possibly limiting the variability needed for stronger behavioral inference.

From Model to Prevention: Strategic Implications

Key Predictors and Insights:

- Perceived Behavior Control & Injunctive Norms = Strongest Smoking Avoidance Predictor.
- Peer-led interventions > Traditional Media Campaigns.

Strategic Actions and Inferences:

- Leverage Public Data = Cost-effective.
- Real-world model to promote smoke-free environments.
- Apply Multi-Level Approach:
  - Empowers Peers.
  - Reinforces Behavior Control
  - Transforms Social Norms.

✓ A systems thinking perspective makes the model scalable, culturally responsive, and cost-effective.

Competencies Demonstrated Through this Master's Thesis

Core Competencies:

1. MPH 3. Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming, and software, as appropriate.
2. MPH 4. Interpret results of data analysis for public health research, policy, or practice.
3. MPH 22. Apply systems thinking tools to a public health issue.

Concentration Competencies:

1. MPH BSTP 3. Apply advanced (multivariate) descriptive and inferential techniques used with public health data.

3. MPH BSTP 8. Interpret results of statistical analyses found in public health studies.

Acknowledgments

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- **My family** – For your constant support, love, and belief in me every step of the way. Your presence has been my foundation.

Supplementary Materials

View the interactive Quarto website for full thesis details and data visualizations, including the embedded R Shiny app:

- https://myechow.github.io/myechow/

✓ Digital tools enhance accessibility, transparency, and public communication of research findings.

Questions

Thank you for your attention. Please feel free to ask any questions or share feedback.